

SOFTWARE ARCHITECTURE DOCUMENT (SAD)

UCE-BusLink

Intelligent University Night Transportation System

Version:	3.0 Enterprise
Date:	May 2026
Classification:	Technical Architecture Specification
Responsible:	Lead Software Architect
Status:	Ready for Development

SAD v2.0 + SAD v2.5 Consolidation

Contents

1	System Overview	6
1.1	Problem Description	6
1.2	Architectural Objectives	6
1.3	System Components	6
1.3.1	Clients	6
1.3.2	Backend	7
2	Monorepo Strategy	8
2.1	Monorepo Justification	8
2.1.1	Advantages of the Monorepo for the Team	8
2.2	Complete Monorepo Structure	8
2.2.1	uce-buslink-backend/ Directory	9
2.2.2	uce-buslink-mobile/ Directory	10
2.2.3	uce-buslink-desktop/ Directory	10
2.3	Monorepo Workflow	10
2.3.1	Local Development	10
2.3.2	Continuous Integration	10
3	Detailed Backend Structure (Maven Multi-Module)	11
3.1	Complete Backend Directory Tree	11
3.2	Maven Configuration Rules	12
3.2.1	The Parent POM is of type pom	12
3.2.2	Dependency Hierarchy Between Modules	13
3.2.3	Flyway Configuration	13
4	Technology Stack Justification	14
4.1	Backend: Spring Boot (Java)	14
4.1.1	Thread Management and Concurrency	14
4.1.2	ACID Transactions and Data Integrity	14
4.2	Mobile Frontend: React Native	15
4.2.1	Access to Background Geolocation APIs	15
4.2.2	Map Rendering Performance	15
4.3	Desktop Application: Electron + React	16
4.3.1	Use Case and Scope	16
4.3.2	Advantages of Electron	16

4.4	Database: PostgreSQL	16
4.4.1	Justification of the Relational Model	16
4.4.2	PostGIS Extension for Geospatial Queries	16
4.5	Cache and Real Time: Redis	16
4.5.1	Decoupling GPS Ingestion from the Database	16
5	Mapping Bounded Contexts (DDD) to Modules	18
5.1	Identity Module (Authentication and User Management)	18
5.1.1	Responsibilities	18
5.1.2	Domain Entities	18
5.1.3	Application Services	18
5.2	Reservations Module (Reservation and Seat Management)	19
5.2.1	Responsibilities	19
5.2.2	Domain Entities	19
5.2.3	Domain Aggregate – Reservation Aggregate	19
5.3	Tracking Module (Real-Time Geolocation)	19
5.3.1	Responsibilities	19
5.4	Notifications Module (Push Notifications and Alerts)	19
5.4.1	Responsibilities	19
5.5	Supervisor Module (Central Monitoring)	20
5.5.1	Responsibilities	20
6	Formal Interface Contract (REST API and WebSocket)	21
6.1	General Conventions	21
6.2	Role-Based Access Control Matrix (RBAC)	21
6.2.1	Role and Permission Definitions	21
6.2.2	Validation in Spring Security	22
6.3	Authentication Module: <code>/api/v1/auth</code>	23
6.3.1	POST <code>/api/v1/auth/google-login</code>	23
6.3.2	POST <code>/api/v1/auth/refresh</code>	25
6.3.3	POST <code>/api/v1/auth/logout</code>	27
6.4	Reservations Module: <code>/api/v1/reservations</code>	27
6.4.1	POST <code>/api/v1/reservations</code>	27
6.4.2	GET <code>/api/v1/reservations</code>	31
6.4.3	DELETE <code>/api/v1/reservations/{id}</code>	33
6.5	Routes and Stops Module: <code>/api/v1/routes</code>	35
6.5.1	GET <code>/api/v1/routes</code>	35
6.5.2	GET <code>/api/v1/routes/{routeId}/trips</code>	36
6.6	Tracking Module: <code>/api/v1/tracking</code> and WebSocket	37
6.6.1	GET <code>/api/v1/tracking/trips/{tripId}/location</code>	37
6.6.2	WebSocket: <code>wss://api.buslink.uce.edu.ec/ws/tracking</code>	38
6.7	Users Module: <code>/api/v1/users</code>	41
6.7.1	GET <code>/api/v1/users/profile</code>	41

6.7.2	GET /api/v1/users/reservations/history	43
6.8	Drivers Module: /api/v1/drivers	45
6.8.1	GET /api/v1/drivers/assigned-trip	45
6.8.2	PATCH /api/v1/trips/{tripId}/boarding	47
6.9	Notifications Module: /api/v1/notifications	50
6.9.1	POST /api/v1/notifications/register-device	50
6.9.2	PUT /api/v1/notifications/preferences	51
6.10	Administration Module: /api/v1/admin	52
6.10.1	POST /api/v1/admin/routes	52
6.10.2	POST /api/v1/admin/trips	54
6.10.3	PATCH /api/v1/admin/users/{id}/trust-score	55
6.11	Supervision Module: /api/v1/supervisor	56
6.11.1	GET /api/v1/supervisor/all-locations	56
6.11.2	GET /api/v1/supervisor/statistics	58
6.11.3	GET /api/v1/supervisor/incidents	59
6.11.4	POST /api/v1/supervisor/incidents/{id}/note	60
6.11.5	POST /api/v1/supervisor/trips/{id}/cancel	61
6.11.6	GET /api/v1/supervisor/audit-logs	62
6.12	Global Error Handling and HTTP Codes	64
6.12.1	Standardized HTTP Codes	64
6.12.2	Standard Error Structure	64
6.12.3	Rate Limit Headers	65
6.13	WebSocket: Supervisor Channel (All-Buses Broadcast)	65
6.13.1	Subscription to Supervisor Channel	65
6.14	Pagination Conventions	67
6.15	Recommended Security Headers	68
7	Data Model (Entity-Relationship)	69
7.1	Primary Entities	69
7.1.1	User	69
7.1.2	Route	69
7.1.3	Stop	69
7.1.4	Bus	70
7.1.5	Trip	70
7.1.6	Seat	70
7.1.7	Reservation	70
7.2	Critical Indexes for Performance	71
8	Persistence Strategy and Migrations	72
8.1	Schema Versioning with Flyway	72
8.1.1	Problem with Hibernate DDL-Auto in Production	72
8.1.2	Flyway as Project Standard	72
8.1.3	Defined Migration Scripts	73

8.2	JPA and Entity Mapping Configuration	73
8.2.1	N+1 Query Problem	73
8.2.2	Solution: Lazy Loading + Explicit Fetch	73
9	Resilience and Reconnection Protocols	74
9.1	Context: Connectivity in Quito	74
9.2	Heartbeat: Zombie Connection Detection	74
9.2.1	Server Mechanism (Spring WebFlux)	74
9.2.2	Client Mechanism (React Native)	74
9.3	Reconnection with Exponential Backoff	75
9.4	Local GPS Coordinate Queue	75
10	Security Matrix	76
10.1	CORS Configuration	76
10.1.1	Allowed Origin Groups	76
10.2	SSL/TLS	76
10.3	Rate Limiting	76
11	Development Schedule	77

Chapter 1

System Overview

1.1 Problem Description

The Central University of Ecuador (UCE) operates a night transportation service for students and staff. The current system lacks real-time visibility, efficient reservation management, and trust mechanisms between users. UCE-BusLink solves these problems through an integrated technological platform.

1.2 Architectural Objectives

The fundamental architectural objectives are:

- Guarantee no seat reservation overbooking under high concurrency.
- Provide real-time GPS tracking with latency below 50ms.
- Support exclusive institutional authentication for @uce.edu.ec domains.
- Scale to 2,000 simultaneous WebSocket connections without degradation.
- Maintain 24/7 availability with network fault tolerance.

1.3 System Components

The system consists of four clients and one centralized backend:

1.3.1 Clients

- **Student Mobile Application** (React Native): seat reservation, real-time bus visualization.
- **Driver Mobile Application** (React Native): GPS transmission, boarding confirmation.

- **Supervisor Desktop Application** (Electron + React): centralized monitoring of all buses simultaneously.
- **Administrative Web Application** (optional/future): route, bus, and user management.

1.3.2 Backend

- **Centralized REST API** (Spring Boot): business logic, authentication, and reservation management.
- **WebSocket Server** (Spring WebFlux): real-time GPS location distribution.
- **Real-Time Cache** (Redis): temporary GPS coordinate storage.
- **Transactional Database** (PostgreSQL): data persistence with ACID guarantees.
- **Notification Service** (Firebase Cloud Messaging): push notifications to devices.

Chapter 2

Monorepo Strategy

2.1 Monorepo Justification

UCE-BusLink will adopt a monorepo architecture, which consists of maintaining the entire ecosystem within the same GitHub repository while completely separating technologies at the root level. This is the most suitable option for academic or short-term projects because with a single `git clone`, the entire team immediately gains access to the whole system.

2.1.1 Advantages of the Monorepo for the Team

- **Simplified DevOps:** DevOps configures pipelines in a single location (`.github/workflows/`). If changes are detected in the backend folder, deployment to AWS is triggered; if changes are detected in mobile, Expo compilation is triggered.
- **Immediate Synchronization:** If an API contract is updated in the backend, the frontend team immediately sees it in the adjacent folder and can update their Axios interceptors without needing to pull changes from another repository.
- **Unified Version Control:** The entire system evolves under the same commit history, facilitating change traceability and team coordination.
- **Single Configuration:** One `.gitignore` file excludes both Java `/target` folders and JavaScript `/node_modules`.

2.2 Complete Monorepo Structure

The root of the UCE-BusLink repository is structured as follows, clearly separating backend, mobile frontend, and desktop application:

```
1 uce-buslink/ (Single Repository Root)
2   .github/
```



```
3         workflows/
4             backend-deploy.yml
5             mobile-build.yml
6             desktop-build.yml
7             pr-validation.yml
8     uce-buslink-backend/
9         uce-buslink-shared-kernel/
10        uce-buslink-identity-module/
11        uce-buslink-reservations-module/
12        uce-buslink-tracking-module/
13        uce-buslink-notifications-module/
14        uce-buslink-supervisor-module/
15        uce-buslink-main/
16    docker/
17        docker-compose.local.yml
18        Dockerfile.backend
19        pom.xml (Parent POM)
20    uce-buslink-mobile/
21        src/
22            screens/
23            components/
24            hooks/
25            services/
26            store/
27            utils/
28        package.json
29    uce-buslink-desktop/
30        src/
31            renderer/
32            main/
33            preload/
34        package.json
35    docs/
36        architecture/
37        api-contracts/
38        diagrams/
39    .gitignore
40    README.md
```

Listing 2.1: UCE-BusLink Monorepo Structure

2.2.1 uce-buslink-backend/ Directory

Contains the ENTIRE Maven multi-module structure for Spring Boot. This folder is self-sufficient and can be compiled independently using `mvn clean package`. It includes the

domain modules (identity, reservations, tracking, notifications, supervisor), the shared-kernel, and the main module that starts the application.

2.2.2 uce-buslink-mobile/ Directory

Independent folder for the React Native Mobile Frontend. It contains views, reusable components, custom hooks (such as `useGPSTracking` for geolocation and WebSocket management), API services, global state management (Redux or Zustand), UCE institutional assets, and Expo or React Native CLI configuration.

2.2.3 uce-buslink-desktop/ Directory

Independent folder for the Monitoring Center in Electron. It contains React components for the supervisor interface, Electron native code (Main Process) for operating system access, the `preload.js` file acting as a secure bridge (contextBridge) for local SQLite access, and local database schemas for offline persistence.

2.3 Monorepo Workflow

2.3.1 Local Development

Each developer clones the complete repository:

```
1 git clone https://github.com/uce/uce-buslink.git
2 cd uce-buslink
```

Then, depending on their work area:

- **Backend:** `cd uce-buslink-backend && mvn spring-boot:run`
- **Mobile:** `cd uce-buslink-mobile && npm install && npm start`
- **Desktop:** `cd uce-buslink-desktop && npm install && npm run dev`

2.3.2 Continuous Integration

GitHub Actions workflows (`.github/workflows/`) automatically detect which part of the system changed and execute only the corresponding pipelines. For example, if a commit only modifies files in `uce-buslink-mobile/`, only `mobile-build.yml` is executed, avoiding unnecessary backend builds.

Chapter 3

Detailed Backend Structure (Maven Multi-Module)

The UCE-BusLink backend is organized as a modular monolith using Maven Multi-Module. Each module represents a Domain-Driven Design Bounded Context with unique responsibilities and explicit boundaries.

3.1 Complete Backend Directory Tree

```
1 uce-buslink-backend/  
2     .github/  
3         workflows/  
4             staging-deploy.yml  
5             pr-validation.yml  
6     docker/  
7         docker-compose.local.yml  
8         Dockerfile.backend  
9     uce-buslink-shared-kernel/  
10         src/main/java/com/ucebuslink/shared/  
11             dto/  
12             exception/  
13             vo/  
14             constant/  
15         pom.xml  
16     uce-buslink-identity-module/  
17         src/main/java/com/ucebuslink/identity/  
18             domain/  
19             application/  
20             infrastructure/  
21             web/  
22         pom.xml  
23     uce-buslink-reservations-module/
```

```

24         src/main/java/com/ucebuslink/reservations/
25         pom.xml
26     uce-buslink-tracking-module/
27         src/main/java/com/ucebuslink/tracking/
28         pom.xml
29     uce-buslink-notifications-module/
30         src/main/java/com/ucebuslink/notifications/
31         pom.xml
32     uce-buslink-supervisor-module/
33         src/main/java/com/ucebuslink/supervisor/
34         pom.xml
35     uce-buslink-main/
36         src/main/java/com/ucebuslink/main/
37         config/
38         BusLinkApplication.java
39     src/main/resources/
40         db/migration/
41             V1__Initial_schema.sql
42             V2__Add_trust_score_table.sql
43             V3__Create_location_history_table.sql
44             V4__Add_device_tokens_and_preferences.
45
46     sql
47
48         application.properties
49         application-dev.properties
50         application-staging.properties
51         application-prod.properties
52
53     pom.xml
54
55     .gitignore
56     pom.xml (PARENT POM)

```

Listing 3.1: Backend Structure

3.2 Maven Configuration Rules

3.2.1 The Parent POM is of type pom

The `pom.xml` file at the root does not package code; its sole purpose is to define inheritance for global dependencies (Java version 17 or 21, Spring Boot Starter Parent) and list submodules within the `<modules>` tag. This allows all components to be compiled in the correct dependency order with a single command (`mvn clean package`).

3.2.2 Dependency Hierarchy Between Modules

To guarantee the Global Isolation Rules, dependencies between modules strictly follow this hierarchical order:

1. **Shared Kernel** depends on no one. It is completely agnostic and only contains DTOs, Value Objects, and base exceptions.
2. **Identity, Reservations, Tracking, Notifications, and Supervisor** depend only on the Shared Kernel. They cannot have cross-dependencies between each other.
3. **Main** depends on all previous modules. As the final packager, it combines them into a single executable `.jar` file.

3.2.3 Flyway Configuration

The `flyway-core` dependency must be declared in the `uce-buslink-main` module, where the physical resources reside (`src/main/resources/db/migration`). This way, when the Docker container or AWS instance executes the application, migrations are triggered before Spring Boot initializes network connections.

Chapter 4

Technology Stack Justification

4.1 Backend: Spring Boot (Java)

4.1.1 Thread Management and Concurrency

UCE-BusLink requires processing multiple simultaneous requests from students attempting to reserve seats on the same bus. The choice of Spring Boot over alternatives such as Node.js is based on its native multi-threaded model. Spring Boot on the JVM uses Tomcat as an application server with a configurable thread pool. Each HTTP request obtains its own worker thread, allowing:

- Blocking database operations without impacting other clients.
- Truly parallel processing on multi-core systems.
- Natural horizontal scaling.

Java provides native synchronization primitives such as `synchronized`, `ReentrantLock`, and `Semaphore`, which guarantee atomicity in critical read-modify-write operations on available seats. These are guarantees of the language and platform, not abstractions built on top of them.

4.1.2 ACID Transactions and Data Integrity

The Reservations module is critical: overbooking cannot be allowed under any circumstances. Spring Boot with Hibernate/JPA provides fine-grained control over transaction isolation levels:

- **READ_COMMITTED** (default): prevents dirty reads.
- **REPEATABLE_READ**: guarantees that seat count readings do not change during the transaction.
- **SERIALIZABLE**: maximum consistency, used in reservation operations under high contention.

Locking Mechanisms

Pessimistic Locking: When starting a reservation transaction, an exclusive lock is acquired on the trip row in PostgreSQL. While this lock is active, any other transaction attempting to modify the same trip is blocked and must wait. This mechanism is recommended for high-contention scenarios such as university rush hours.

Optimistic Locking (with @Version): Alternative for low-contention scenarios. Each entity includes a version field incremented on every UPDATE. If two transactions attempt to modify the same row simultaneously, the second detects that the version changed and throws an exception, forcing a retry.

4.2 Mobile Frontend: React Native

4.2.1 Access to Background Geolocation APIs

The most critical limitation of Progressive Web Apps (PWA) for this system is the inability to reliably maintain active geolocation in the background. The Web Geolocation API only works while the tab is in the foreground. On iOS, Apple restricts Service Workers to a maximum of 10 minutes in the background.

Advantages of React Native:

- On iOS, `CLLocationManager` with `requestAlwaysAndWhenInUseAuthorization()` allows location updates even when the screen is off.
- On Android, `FusedLocationProvider` with `PRIORITY_HIGH_ACCURACY` maintains updates every 1-5 seconds in the background.
- Native APIs automatically optimize battery consumption.

4.2.2 Map Rendering Performance

During rush hour, there may be 150 or more students simultaneously waiting for buses. Load testing on maps shows that with PWAs using Leaflet or Mapbox in a web browser, framerate drops to 15-20 FPS once 200 markers are exceeded, resulting in a laggy experience. With React Native using `react-native-maps`, which wraps native MapKit and Google Maps, 50-60 FPS can be maintained with more than 200 markers because the operating system's map engine handles rendering.

4.3 Desktop Application: Electron + React

4.3.1 Use Case and Scope

The UCE Monitoring Center is a desktop application deployed on university operator workstations. Its purpose is to supervise in real time all buses operating simultaneously during the night shift. The fundamental difference compared to the mobile app is that the student visualizes only one bus (the one reserved), while the supervisor visualizes all buses plus analytics and incident management.

4.3.2 Advantages of Electron

- Local installation without requiring additional web server infrastructure.
- Offline-first support with historical storage in local SQLite.
- Integration with the operating system for native notifications, disk access, and desktop application behavior (shortcuts, hotkeys, tray icon).
- Enhanced security with certificate pinning and stricter server validation.

4.4 Database: PostgreSQL

4.4.1 Justification of the Relational Model

The business model requires relationships and referential integrity naturally expressed in a relational model. PostgreSQL provides referential integrity through foreign keys, guaranteeing that orphan reservations, seats without trips, or trips without buses cannot exist. Unique constraints such as `UNIQUE(trip_id, seat_number)` guarantee at the database level that two seats with the same number cannot exist on the same trip.

4.4.2 PostGIS Extension for Geospatial Queries

PostgreSQL with the PostGIS extension allows storing geographic points and executing proximity queries, such as finding stops within 500 meters of the user's location. This is critical for ETA (estimated time of arrival) calculations and nearby stop search functionality.

4.5 Cache and Real Time: Redis

4.5.1 Decoupling GPS Ingestion from the Database

The driver emits GPS location updates every 2 seconds. With 10 simultaneously active buses, writing directly to PostgreSQL would generate approximately 720,000 writes per

day, producing disk I/O overhead, contention in the location table, and exponential historical storage growth. Redis acts as a real-time read buffer with a TTL of 60 seconds, protecting PostgreSQL from constant writes while maintaining latency below 5ms for reads.

Chapter 5

Mapping Bounded Contexts (DDD) to Modules

The backend is organized following Domain-Driven Design (DDD) principles, where each module represents a Bounded Context with unique responsibilities and explicit boundaries.

5.1 Identity Module (Authentication and User Management)

5.1.1 Responsibilities

- User validation against Google OAuth 2.0
- JWT issuance and renewal
- Role management
- @uce.edu.ec domain validation
- Refresh token registration and revocation

5.1.2 Domain Entities

User and Token (to model refresh tokens with individual or full-user revocation).

5.1.3 Application Services

- **AuthenticationService:** `authenticateWithGoogle`, `refreshToken`, `validateJWT`, `logout`
- **UserService:** `getUserById`, `updateUserProfile`, `getRolesByUserId`

5.2 Reservations Module (Reservation and Seat Management)

5.2.1 Responsibilities

- Management of trips, routes, stops, and buses
- Seat reservation with guaranteed no overbooking
- Reservation cancellation
- Availability queries

5.2.2 Domain Entities

Route, Stop, Bus, Trip (with version field for locking), Seat (with UNIQUE constraint by trip and number, version field), Reservation (with UNIQUE constraint by trip and user, version field).

5.2.3 Domain Aggregate – Reservation Aggregate

- **Aggregate Root:** Reservation
- **Emitted events:** `ReservationCreatedEvent` (consumed by Notifications Module and Trust Module), `ReservationCancelledEvent` (consumed by Notifications Module)

5.3 Tracking Module (Real-Time Geolocation)

5.3.1 Responsibilities

- GPS coordinate ingestion from drivers via WebSocket
- Temporary storage in Redis
- Distribution of locations to subscribed students
- ETA calculation for upcoming stops

5.4 Notifications Module (Push Notifications and Alerts)

5.4.1 Responsibilities

- Registration of FCM tokens for mobile devices
- Sending push notifications

- Management of user notification preferences

5.5 Supervisor Module (Central Monitoring)

5.5.1 Responsibilities

- Real-time aggregation of the status of all buses
- Global system statistics
- Incident management
- Audit logs
- Dedicated WebSocket channel for the Electron application

Chapter 6

Formal Interface Contract (REST API and WebSocket)

6.1 General Conventions

- **Base URL:** `https://api.buslink.uce.edu.ec/api/v1`
- **Authentication:** Header `Authorization: Bearer {accessToken}` on all endpoints except those marked as public
- **Format:** `Content-Type: application/json` on all requests and responses
- **Pagination:** Listing endpoints accept the parameters `page` (default: 1) and `size` (default: 20, maximum: 100)
- **Rate Limiting:** Specified per endpoint; exceeding the limit returns HTTP 429 Too Many Requests

6.2 Role-Based Access Control Matrix (RBAC)

6.2.1 Role and Permission Definitions

Role: STUDENT

Permissions:

```
create:reservation (POST /api/v1/reservations)
read:own_reservation (GET /api/v1/reservations)
delete:own_reservation (DELETE /api/v1/reservations/{id})
read:own_tracking (WebSocket /ws/tracking, subscription to own trip)
read:own_profile (GET /api/v1/users/profile)
read:own_history (GET /api/v1/users/reservations/history)
read:own_trust_score (GET /api/v1/users/profile)
read:routes (GET /api/v1/routes)
read:route_trips (GET /api/v1/routes/{routeId}/trips)
```

Role: DRIVER

Permissions:

```
write:gps_update (WebSocket /ws/tracking, location updates)
read:assigned_trip (GET /api/v1/drivers/assigned-trip)
write:boarding_confirmation (PATCH /api/v1/trips/{tripId}/boarding)
read:own_profile (GET /api/v1/users/profile)
read:routes (GET /api/v1/routes)
read:assigned_trip_details (GET /api/v1/drivers/assigned-trip)
```

Role: ADMIN

Permissions:

```
create:route (POST /api/v1/admin/routes)
update:route (PUT /api/v1/admin/routes/{id})
delete:route (DELETE /api/v1/admin/routes/{id})
create:trip (POST /api/v1/admin/trips)
update:trip (PUT /api/v1/admin/trips/{id})
cancel:trip (POST /api/v1/admin/trips/{id}/cancel)
create:bus (POST /api/v1/admin/buses)
update:bus (PUT /api/v1/admin/buses/{id})
read:all_users (GET /api/v1/admin/users)
write:trust_score (PATCH /api/v1/admin/users/{id}/trust-score)
suspend:user (POST /api/v1/admin/users/{id}/suspend)
read:audit_logs (GET /api/v1/admin/audit-logs)
```

Role: SUPERVISOR (Electron Application - Monitoring Center)

Permissions:

```
read:all_trips_real_time (WebSocket /ws/tracking, all-buses channel)
read:all_locations (GET /api/v1/supervisor/all-locations)
read:all_statistics (GET /api/v1/supervisor/statistics)
read:incident_reports (GET /api/v1/supervisor/incidents)
write:trip_cancellation (POST /api/v1/supervisor/trips/{id}/cancel)
read:audit_logs (GET /api/v1/supervisor/audit-logs)
read:all_users (GET /api/v1/supervisor/users)
write:incident_note (POST /api/v1/supervisor/incidents/{id}/note)
```

6.2.2 Validation in Spring Security

All endpoints are protected with `@PreAuthorize` decorators:

```
@RestController
@RequestMapping("/api/v1")
public class ApiController {
```

```
@PostMapping("/reservations")
@PreAuthorize("hasRole('STUDENT')")
public ResponseEntity<...> createReservation(...) { ... }

@GetMapping("/supervisor/all-locations")
@PreAuthorize("hasRole('SUPERVISOR')")
public ResponseEntity<...> getAllLocations(...) { ... }
}
```

If the JWT does not contain the required role:

→ `HttpStatus.FORBIDDEN` (403)

6.3 Authentication Module: /api/v1/auth

6.3.1 POST /api/v1/auth/google-login

Purpose: Authenticate a user with Google OAuth 2.0 and obtain a custom JWT.

Authentication: Public.

Rate Limit: 10 requests per minute per IP.

Request

POST /api/v1/auth/google-login HTTP/1.1

Host: api.buslink.uce.edu.ec

Content-Type: application/json

```
{
  "googleIdToken": "eyJhbGc...",
  "googleAccessToken": "ya29..."
}
```

VALIDATIONS:

- googleIdToken: cannot be empty; must be a valid JWT (3 parts separated by .)
- googleAccessToken: cannot be empty; range 200-1000 characters
- Email must end with @uce.edu.ec

Response 200 OK - Success

HTTP/1.1 200 OK

Content-Type: application/json

```
{
```

```
"accessToken": "eyJhbGciOiJIUzI1NiJ9...",
"refreshToken": "opaque_hash_token_abc123...",
"expiresIn": 900,
"usuario": {
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "email": "juan.garcia@uce.edu.ec",
  "nombres": "Juan",
  "apellidos": "García Martínez",
  "rol": "STUDENT"
}
```

FIELDS:

- accessToken: JWT with claims {sub (userId), email, role, iat, exp}
- refreshToken: opaque token (not decodable by client); store in AsyncStorage
- expiresIn: 900 seconds (15 minutes) until accessToken expiration
- usuario: basic User object

Response 400 Bad Request

HTTP/1.1 400 Bad Request

Content-Type: application/json

```
{
  "error": "invalid_request",
  "error_description": "googleIdToken is required and must be a valid JWT",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

CASES:

- Empty or malformed googleIdToken
- Empty googleAccessToken
- Malformed JSON request

Response 401 Unauthorized

HTTP/1.1 401 Unauthorized

Content-Type: application/json

```
{
  "error": "invalid_token",
  "error_description": "Google ID token signature is invalid or expired",
  "timestamp": "2026-05-20T10:30:00Z"
}
```



```
}
```

CASES:

- Expired Google token
- Invalid signature (tampering)
- Audience mismatch (token from another app)
- Google API validation rejected

Response 403 Forbidden

HTTP/1.1 403 Forbidden

Content-Type: application/json

```
{
  "error": "invalid_email_domain",
  "error_description": "Only @uce.edu.ec email addresses are allowed",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

CASES:

- Email does not end with @uce.edu.ec

Response 500 Internal Server Error

HTTP/1.1 500 Internal Server Error

Content-Type: application/json

```
{
  "error": "server_error",
  "error_description": "An unexpected error occurred. Please try again later.",
  "timestamp": "2026-05-20T10:30:00Z",
  "traceId": "a1b2c3d4e5f6"
}
```

CASES:

- Database unavailable
- Google API not responding
- Credential configuration error

6.3.2 POST /api/v1/auth/refresh

Purpose: Renew Access Token using a Refresh Token (token rotation).

Rate Limit: 20 requests per minute per IP.

Request

POST /api/v1/auth/refresh HTTP/1.1

Host: api.buslink.uce.edu.ec

Content-Type: application/json

```
{
  "refreshToken": "opaque_hash_token_from_login"
}
```

VALIDATIONS:

- refreshToken: cannot be empty; must exist in DB and not be revoked

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "accessToken": "eyJhbGciOiJIUzI1NiJ9...",
  "refreshToken": "opaque_new_token_xyz789...",
  "expiresIn": 900
}
```

NOTE: The new refreshToken may differ from the previous one (rotation).
Client must update AsyncStorage with the new token.

Response 401 Unauthorized

HTTP/1.1 401 Unauthorized

Content-Type: application/json

```
{
  "error": "invalid_grant",
  "error_description": "Refresh token is invalid, expired, or revoked",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

CASES:

- Token does not exist in DB
- Token.revoked = true
- Token.expiration < now
- Token tampered with

→ Client must navigate to LoginScreen

6.3.3 POST /api/v1/auth/logout

Purpose: Invalidate Refresh Token (revoke session).

Authentication: Bearer token required.

Rate Limit: 5 requests per minute per user.

Request

POST /api/v1/auth/logout HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Content-Type: application/json

```
{
  "refreshToken": "opaque_token (optional)"
}
```

NOTE:

- If refreshToken is provided: only that token is invalidated (specific device logout)
- If omitted: invalidates all user refresh tokens (global logout)

Response 204 No Content

HTTP/1.1 204 No Content

(no body)

Response 401 Unauthorized

HTTP/1.1 401 Unauthorized

Content-Type: application/json

```
{
  "error": "unauthorized",
  "error_description": "Access token is invalid or expired"
}
```

6.4 Reservations Module: /api/v1/reservations

6.4.1 POST /api/v1/reservations

Purpose: Create a new seat reservation with ACID guarantees to prevent overbooking.

Authentication: Bearer token required (role: STUDENT).

Rate Limit: 5 requests per minute per user.

Request

POST /api/v1/reservations HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Content-Type: application/json

```
{
  "tripId": "550e8400-e29b-41d4-a716-446655440000",
  "preferredSeat": 5,
  "boardingStop": "650e8400-e29b-41d4-a716-446655441111"
}
```

VALIDATIONS:

- tripId: cannot be null; must be a valid UUID
- preferredSeat: if provided, must be > 0 and <= bus capacity
- boardingStop: if provided, must exist in the trip route stops

BACKEND PROCESSING:

1. Validate JWT and role == STUDENT
2. Retrieve Trip with PESSIMISTIC LOCK (guarantee atomicity)
3. Verify trip status (SCHEDULED or ONGOING)
4. Count available seats
5. If preferredSeat specified: validate availability
6. If not specified: automatically assign the first available seat
7. Change Seat status to RESERVED
8. Create Reservation record
9. Emit ReservationCreatedEvent
10. COMMIT transaction (lock release)

Response 201 Created - Success

HTTP/1.1 201 Created

Location: /api/v1/reservations/760e8400-e29b-41d4-a716-446655442222

Content-Type: application/json

```
{
  "id": "760e8400-e29b-41d4-a716-446655442222",
  "tripId": "550e8400-e29b-41d4-a716-446655440000",
  "seatId": "850e8400-e29b-41d4-a716-446655443333",
}
```

```
"seatNumber": 5,
"studentId": "950e8400-e29b-41d4-a716-446655444444",
"status": "ACTIVE",
"createdAt": "2026-05-20T10:30:00Z",
"bus": {
  "id": "050e8400-e29b-41d4-a716-446655445555",
  "licensePlate": "UIO-1234"
},
"route": {
  "id": "150e8400-e29b-41d4-a716-446655446666",
  "name": "Main Campus Ring Route"
},
"departureTime": "2026-05-20T16:00:00Z"
}
```

Response 400 Bad Request

HTTP/1.1 400 Bad Request
Content-Type: application/json

```
{
  "error": "invalid_input",
  "error_description": "Trip ID must be a valid UUID",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

CASES:

- tripId is not a valid UUID
- tripId is null
- preferredSeat out of range

Response 404 Not Found

HTTP/1.1 404 Not Found
Content-Type: application/json

```
{
  "error": "trip_not_found",
  "error_description": "Trip with ID {tripId} does not exist",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

Response 409 Conflict

HTTP/1.1 409 Conflict

Content-Type: application/json

```
{
  "error": "no_available_seats",
  "error_description": "All seats for this trip are reserved",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

OR (if the user already reserved on this trip):

```
{
  "error": "duplicate_reservation",
  "error_description": "You have already reserved a seat for this trip",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

OR (if the trip already started):

```
{
  "error": "trip_already_started",
  "error_description": "Cannot reserve for a trip that has already started",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

OR (if trust score is insufficient):

```
{
  "error": "insufficient_trust_score",
  "error_description": "Your trust score is too low to reserve at this moment",
  "timestamp": "2026-05-20T10:30:00Z",
  "data": {
    "currentScore": 15,
    "minimumRequired": 50
  }
}
```

OR (if the specific seat was reserved during lock wait):

```
{
```

```
"error": "seat_no_longer_available",
"error_description": "The seat you selected was just reserved by another user",
"timestamp": "2026-05-20T10:30:00Z"
}
```

Response 401 Unauthorized

HTTP/1.1 401 Unauthorized
Content-Type: application/json

```
{
  "error": "unauthorized",
  "error_description": "Access token is invalid or expired"
}
```

Response 500 Internal Server Error

HTTP/1.1 500 Internal Server Error
Content-Type: application/json

```
{
  "error": "database_error",
  "error_description": "Failed to save reservation. Please try again.",
  "timestamp": "2026-05-20T10:30:00Z",
  "traceId": "x1y2z3w4"
}
```

CASES:

- Transaction aborted due to timeout
- Database deadlock (very rare with PESSIMISTIC_WRITE)
- Database error

6.4.2 GET /api/v1/reservations

Purpose: Retrieve active and upcoming reservations for the authenticated user.

Authentication: Bearer token required.

Request

GET /api/v1/reservations?status=ACTIVE&upcomingWeeks=2&sortBy=departure_date_asc HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLT...

QUERY PARAMETERS:

- status: enum [ACTIVE, CANCELLED_BY_STUDENT, CANCELLED_BY_ADMIN, NO_SHOW, ALL]
(default: ACTIVE)
- upcomingWeeks: integer 1-4 (default: 4)
- sortBy: enum [departure_date_asc, departure_date_desc]
(default: departure_date_asc)
- page: integer (default: 1)
- size: integer (default: 20, maximum: 100)

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "total": 5,
  "page": 1,
  "size": 20,
  "reservations": [
    {
      "id": "760e8400-e29b-41d4-a716-446655442222",
      "tripId": "550e8400-e29b-41d4-a716-446655440000",
      "seatNumber": 5,
      "status": "ACTIVE",
      "createdAt": "2026-05-15T10:30:00Z",
      "cancelledAt": null,
      "bus": {
        "id": "050e8400-e29b-41d4-a716-446655445555",
        "licensePlate": "UI0-1234",
        "capacity": 15
      },
      "route": {
        "id": "150e8400-e29b-41d4-a716-446655446666",
        "name": "Main Campus Ring Road"
      },
      "departureTime": "2026-05-20T16:00:00Z",
      "boardingStop": {
        "id": "250e8400-e29b-41d4-a716-446655447777",
        "name": "Main Entrance",
        "order": 1,
        "latitude": -0.2156,
        "longitude": -78.4678
      }
    }
  ]
}
```



```
]
}
```

Response 401 Unauthorized

HTTP/1.1 401 Unauthorized
Content-Type: application/json

```
{
  "error": "unauthorized",
  "error_description": "Access token is invalid or expired"
}
```

6.4.3 DELETE /api/v1/reservations/{id}

Purpose: Cancel a reservation.

Authentication: Bearer token required.

Rate Limit: 5 requests per minute per user.

Request

DELETE /api/v1/reservations/760e8400-e29b-41d4-a716-446655442222 HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...

BACKEND VALIDATIONS:

1. Valid JWT
2. User owns the reservation OR role == ADMIN
3. Reservation.status == ACTIVE
4. Trip has not started OR departure is more than 15 minutes away

IF VALID:

5. If cancellation < 15 min before departure: apply -5 trust points
6. If cancellation >= 15 min before departure: no penalty (0 points)
7. Change status to CANCELLED_BY_STUDENT
8. Release seat (change to AVAILABLE)
9. Emit ReservationCancelledEvent

Response 204 No Content - Success

HTTP/1.1 204 No Content

(no body)

Response 403 Forbidden

HTTP/1.1 403 Forbidden

Content-Type: application/json

```
{
  "error": "not_owner",
  "error_description": "You can only cancel your own reservations (or you are not an admin)",
}
```

CASES:

- User attempts to cancel another user's reservation
- User is not ADMIN

Response 404 Not Found

HTTP/1.1 404 Not Found

Content-Type: application/json

```
{
  "error": "reservation_not_found",
  "error_description": "Reservation with ID {id} does not exist"
}
```

Response 409 Conflict

HTTP/1.1 409 Conflict

Content-Type: application/json

```
{
  "error": "cannot_cancel",
  "error_description": "Reservation cannot be cancelled because the trip has already started",
  "timestamp": "2026-05-20T10:30:00Z",
  "data": {
    "minutesUntilDeparture": 5,
    "minimumMinutesRequired": 15
  }
}
```

OR (if it is already cancelled):

```
{
  "error": "already_cancelled",
  "error_description": "This reservation is already cancelled"
}
```

```
}
```

6.5 Routes and Stops Module: /api/v1/routes

6.5.1 GET /api/v1/routes

Purpose: Retrieve all available routes.

Authentication: Bearer token required.

Request

```
GET /api/v1/routes?active=true&search=Ring%20Road&page=1&size=20 HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...
```

QUERY PARAMETERS:

- active: boolean (default: true; only routes with scheduled trips)
- search: string (search by route name)
- page: integer (default: 1)
- size: integer (default: 20, maximum: 100)

Response 200 OK

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "total": 8,
  "page": 1,
  "size": 20,
  "routes": [
    {
      "id": "150e8400-e29b-41d4-a716-446655446666",
      "name": "Main Campus Ring Road",
      "description": "Night route around the campus ring road",
      "active": true,
      "stops": [
        {
          "id": "250e8400-e29b-41d4-a716-446655447777",
          "name": "Main Entrance",
          "order": 1,
          "latitude": -0.2156,
          "longitude": -78.4678
        }
      ]
    }
  ]
}
```

```
    },
    {
      "id": "350e8400-e29b-41d4-a716-446655448888",
      "name": "Central Library",
      "order": 2,
      "latitude": -0.2170,
      "longitude": -78.4690
    }
  ],
  "totalTrips": 12
}
]
```

6.5.2 GET /api/v1/routes/{routeId}/trips

Purpose: Retrieve available trips for a specific route.

Authentication: Bearer token required.

Request

```
GET /api/v1/routes/150e8400-e29b-41d4-a716-446655446666/trips?
date=2026-05-20&withAvailableSeats=true HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...
```

QUERY PARAMETERS:

- date: string (YYYY-MM-DD; default: next 7 days)
- withAvailableSeats: boolean (default: true)

Response 200 OK

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "trips": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440000",
      "date": "2026-05-20",
      "departureTime": "16:00",
      "estimatedArrivalTime": "17:15",
      "bus": {
```

```
    "id": "050e8400-e29b-41d4-a716-446655445555",
    "licensePlate": "UI0-1234",
    "capacity": 15
  },
  "availableSeats": 7,
  "totalSeats": 15,
  "driver": {
    "id": "450e8400-e29b-41d4-a716-446655449999",
    "name": "Juan Pérez García"
  },
  "status": "SCHEDULED"
}
]
}
```

6.6 Tracking Module: /api/v1/tracking and WebSocket

6.6.1 GET /api/v1/tracking/trips/{tripId}/location

Purpose: Retrieve the current bus location (REST alternative to WebSocket).

Authentication: Bearer token required.

Request

```
GET /api/v1/tracking/trips/550e8400-e29b-41d4-a716-446655440000/location HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...
```

Response 200 OK

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "busId": "050e8400-e29b-41d4-a716-446655445555",
  "tripId": "550e8400-e29b-41d4-a716-446655440000",
  "latitude": -0.2156,
  "longitude": -78.4678,
  "accuracy": 8.5,
  "velocity": 45.2,
  "timestamp": 1234567890000,
  "etaMinutes": 12,
  "nextStop": {
```

```
    "id": "350e8400-e29b-41d4-a716-446655448888",
    "name": "Central Library",
    "order": 2,
    "distanceMeters": 450
  }
}
```

Response 404 Not Found

HTTP/1.1 404 Not Found

Content-Type: application/json

```
{
  "error": "trip_not_found_or_inactive",
  "error_description": "Trip {tripId} not found or is not currently active"
}
```

6.6.2 WebSocket: wss://api.buslink.uce.edu.ec/ws/tracking

Purpose: Bidirectional connection to transmit and receive real-time GPS locations.

Handshake

```
GET /ws/tracking HTTP/1.1
Host: api.buslink.uce.edu.ec
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Key: dGh1IHhnbXBsZSBub25jZQ==
Sec-WebSocket-Version: 13
Authorization: Bearer eyJhbGc...
```

BACKEND VALIDATIONS:

1. Extract JWT from Authorization header
2. Validate signature and expiration
3. Extract role from JWT
4. If role == DRIVER: authorize GPS writing
5. If role == STUDENT: authorize reading (subscription)
6. If role == SUPERVISOR: authorize reading for all buses

RESPONSE:

```
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
```

Sec-WebSocket-Accept: s3pPLMBiTxaQ9kYGzzhZRbK+x0o=

GPS Update Message (Driver → Backend)

Direction: Outgoing (client → server)

Frequency: Every 2 seconds (configurable)

```
{
  "type": "gps_update",
  "payload": {
    "busId": "050e8400-e29b-41d4-a716-446655445555",
    "latitude": -0.2156,
    "longitude": -78.4678,
    "accuracy": 8.5,
    "velocity": 45.2,
    "timestamp": 1234567890000
  }
}
```

VALIDATIONS:

- busId: cannot be empty; must exist in DB
- latitude: range -90 to 90 (decimal degrees)
- longitude: range -180 to 180 (decimal degrees)
- accuracy: must be > 0 (meters)
- velocity: must be >= 0 (km/h)
- timestamp: must be <= now() (milliseconds)
- User.role must be DRIVER
- User must be the assigned driver of busId

BACKEND PROCESSING:

1. Validate JWT and permissions
2. Write to Redis: SET "bus:{busId}:location" {...}
3. Update GEO index in Redis
4. Publish GPSLocationUpdatedEvent
5. Broadcast to /topic/supervisor/all-buses (for SUPERVISOR)
6. Broadcast to /topic/trip/{tripId}/location-update (for STUDENT subscribers)

Subscription Message (Student → Backend)

Direction: Outgoing (client → server)

```
{
  "type": "subscribe_trip_location",
  "tripId": "550e8400-e29b-41d4-a716-446655440000"
}
```

```
}
```

VALIDATIONS:

1. Verify that the user has an ACTIVE reservation for that tripId
2. If not: send error frame and close connection

SUCCESS RESPONSE:

```
{  
  "type": "subscribed",  
  "tripId": "550e8400-e29b-41d4-a716-446655440000"  
}
```

ERROR:

```
{  
  "type": "error",  
  "error": "not_authorized",  
  "message": "You don't have an active reservation for this trip"  
}
```

Location Update Message (Backend → Students)

Direction: Incoming (server → client)

Recipients: All students with an ACTIVE reservation on the trip

```
{  
  "type": "location_update",  
  "data": {  
    "busId": "050e8400-e29b-41d4-a716-446655445555",  
    "tripId": "550e8400-e29b-41d4-a716-446655440000",  
    "latitude": -0.2156,  
    "longitude": -78.4678,  
    "accuracy": 8.5,  
    "velocity": 45.2,  
    "timestamp": 1234567890000,  
    "etaMinutes": 12,  
    "nextStop": {  
      "name": "Central Library",  
      "order": 2,  
      "distanceMeters": 450  
    }  
  }  
}
```


6.7 Users Module: /api/v1/users

6.7.1 GET /api/v1/users/profile

Purpose: Get authenticated user profile plus current Trust Score.

Authentication: Bearer token required.

Rate Limit: 30 requests per minute per user.

Request

GET /api/v1/users/profile HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "usuario": {
    "id": "950e8400-e29b-41d4-a716-446655444444",
    "email": "juan.garcia@uce.edu.ec",
    "nombres": "Juan",
    "apellidos": "García Martínez",
    "rol": "STUDENT",
    "estado": "ACTIVE",
    "fotoPerfil": "https://lh3.googleusercontent.com/a-/...",
    "telefonoContacto": "+593999888777",
    "direccion": "Calle 5, Casa 12, Quito",
    "fechaNacimiento": "2000-05-15",
    "generoCorreo": "juan.garcia",
    "carrera": "Ingeniería en Sistemas",
    "numeroDocumento": "1720456789",
    "fechaPrimerLogin": "2026-01-15T10:30:00Z",
    "fechaUltimoLogin": "2026-05-20T14:45:00Z",
    "fechaCreacionCuenta": "2026-01-10T09:00:00Z"
  },
  "puntuacionConfianza": {
    "id": "a50e8400-e29b-41d4-a716-446655555555",
    "usuarioId": "950e8400-e29b-41d4-a716-446655444444",
    "puntuacion": 85,
    "nivel": "BUENA",
    "reservasTotales": 45,
```

```
"reservasCompletadas": 42,
"noShows": 1,
"cancelacionesUltimo30dias": 2,
"penalizacionesActivas": [
  {
    "id": "b50e8400-e29b-41d4-a716-446655666666",
    "tipo": "LATE_CANCELLATION",
    "razon": "Cancelación a 10 minutos de salida",
    "puntosRestados": 5,
    "fechaInicio": "2026-05-15T10:00:00Z",
    "fechaExpiracion": null,
    "estado": "ACTIVE"
  }
],
"trendultimos7dias": -3,
"proximaRevision": null,
"historico": [
  {
    "fecha": "2026-05-20T14:00:00Z",
    "evento": "Viaje completado",
    "cambio": 2,
    "puntuacionResultante": 85
  },
  {
    "fecha": "2026-05-15T10:00:00Z",
    "evento": "Cancelación tardía (< 15 min)",
    "cambio": -5,
    "puntuacionResultante": 83
  }
]
},
"estadisticas": {
  "viajesTotales": 45,
  "viajesTotalesCompletados": 42,
  "kmTotalesViajados": 287.5,
  "horasTotalesEnTransito": 18.5,
  "diasActivo": 130,
  "tiempoPromedioEspera": 8.3,
  "rutasFrecuentes": [
    {
      "rutaId": "150e8400-e29b-41d4-a716-446655446666",
      "nombreRuta": "Circunvalación Campus Principal",
```

```
        "vecesUtilizada": 25
      }
    ]
  },
  "preferencias": {
    "notificacionesHabilitadas": true,
    "notificacionesEmail": false,
    "compartirUbicacion": true,
    "visibilidadPerfil": "PRIVATE"
  }
}
```

NOTES:

- Trust Score is recalculated on each query if there are pending changes
- Expired penalties do not appear (status = EXPIRED)
- Sensitive data (phone, address) only visible to user or ADMIN

6.7.2 GET /api/v1/users/reservations/history

Purpose: Get reservation history for audit purposes.

Authentication: Bearer token required.

Request

```
GET /api/v1/users/reservations/history?desde=2026-02-20&hasta=2026-05-20&\
estado=COMPLETED&pagina=1&tamano=20&ordenar=fecha_desc HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...
```

QUERY PARAMETERS:

- desde: ISO 8601 date (default: 90 days ago)
- hasta: ISO 8601 date (default: today)
- estado: enum [COMPLETED, CANCELLED_BY_STUDENT, CANCELLED_BY_ADMIN, NO_SHOW, ALL] (default: ALL)
- pagina: integer (default: 1)
- tamano: integer (default: 50, maximum: 500)
- ordenar: enum [fecha_asc, fecha_desc] (default: fecha_desc)

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "total": 42,
  "pagina": 1,
  "tamano": 20,
  "reservacionesHistorico": [
    {
      "id": "760e8400-e29b-41d4-a716-446655442222",
      "tripId": "550e8400-e29b-41d4-a716-446655440000",
      "bus": {
        "id": "050e8400-e29b-41d4-a716-446655445555",
        "numeroPlaca": "UIO-1234"
      },
      "ruta": {
        "id": "150e8400-e29b-41d4-a716-446655446666",
        "nombre": "Circunvalación Campus Principal"
      },
      "asientoNumero": 5,
      "fechaReserva": "2026-05-15T10:30:00Z",
      "horarioSalida": "2026-05-15T16:00:00Z",
      "horarioLlegadaEstimada": "2026-05-15T16:45:00Z",
      "estado": "COMPLETED",
      "detalleEstado": {
        "horaAbordaje": "2026-05-15T15:58:00Z",
        "horaDesabordo": "2026-05-15T16:47:00Z",
        "paradaFinal": {
          "nombre": "Salida Campus",
          "latitud": -0.2160,
          "longitud": -78.4680
        },
        "puntosGanados": 2
      },
      "paradaAbordaje": {
        "nombre": "Entrada Principal",
        "latitud": -0.2156,
        "longitud": -78.4678
      },
      "conductor": {
        "id": "450e8400-e29b-41d4-a716-446655449999",
        "nombre": "Juan Pérez García",
        "calificacion": 4.8
      }
    }
  ]
}
```

```
],
"estadisticasBasadas": {
  "totalCompletados": 42,
  "totalCancelados": 2,
  "totalNoShows": 1,
  "tasaCompletitud": 0.933,
  "tasaNoShow": 0.022,
  "kmPromedio": 6.9
}
}
```

NOTES ON STATUSES:

- COMPLETED: trip completed, user boarded and disembarked
- CANCELLED_BY_STUDENT: user cancelled with notice (≥ 15 min) or late (< 15 min)
- CANCELLED_BY_ADMIN: admin cancelled entire trip
- NO_SHOW: user did not appear within 15 minutes after departure

POINT CALCULATION (immutable in history):

- Completed: +2 points
- Late cancellation: -5 points
- Advance cancellation: 0 points
- No-show: -10 points

6.8 Drivers Module: /api/v1/drivers

6.8.1 GET /api/v1/drivers/assigned-trip

Purpose: Driver queries assigned trip before activating GPS WebSocket.

Authentication: Bearer token required (role: DRIVER).

Rate Limit: 5 requests per minute per driver.

Request

GET /api/v1/drivers/assigned-trip HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

BACKEND VALIDATIONS:

1. Valid JWT
2. User.role == DRIVER
3. User.estado == ACTIVE
4. Assigned trip exists for TODAY with status = SCHEDULED
5. If not exists: return 404

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "viaje": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "bus": {
      "id": "050e8400-e29b-41d4-a716-446655445555",
      "numeroPlaca": "UI0-1234",
      "capacidad": 15,
      "marca": "Hyundai",
      "anio": 2022,
      "kmActuales": 45230,
      "combustibleAproximado": "75% - Full",
      "ultimoMantenimiento": "2026-04-15T00:00:00Z",
      "estado": "OPERATIONAL"
    },
    "ruta": {
      "id": "150e8400-e29b-41d4-a716-446655446666",
      "nombre": "Circunvalación Campus Principal",
      "descripcion": "Ruta que recorre el perímetro del campus principal",
      "paradas": [
        {
          "id": "250e8400-e29b-41d4-a716-446655447777",
          "nombre": "Entrada Principal",
          "orden": 1,
          "latitud": -0.2156,
          "longitud": -78.4678,
          "horarioLlegadaEstimado": "16:00",
          "tiempoDetencionMinutos": 2
        },
        {
          "id": "350e8400-e29b-41d4-a716-446655448888",
          "nombre": "Biblioteca Central",
          "orden": 2,
          "latitud": -0.2170,
          "longitud": -78.4690,
          "horarioLlegadaEstimado": "16:10",
          "tiempoDetencionMinutos": 2
        }
      ]
    }
  }
}
```

```

    ]
  },
  "fecha": "2026-05-20",
  "horaInicio": "16:00",
  "horaFin": "17:15",
  "estado": "SCHEDULED",
  "asientosReservados": 8,
  "asientosTotales": 15,
  "contactoEmergencia": {
    "nombre": "Carlos Operador",
    "telefonoOperador": "+593999111222",
    "emailOperador": "operador@uce.edu.ec"
  },
  "notas": "Ruta con alta demanda. Revisar puntuaciones de confianza de pasajeros."
  "rutaMapaUrl": "https://maps.google.com/maps?q=..."
},
"instrucciones": {
  "pasosPrevios": [
    "Revisar nivel de combustible (mínimo 25%)",
    "Verificar frenos, luces y espejos",
    "Confirmar que asientos están limpios",
    "Activar GPS en aplicación antes de las 15:45",
    "Revisar lista de pasajeros reservados"
  ],
  "protocolo": "Activar geolocalización y conectar a WebSocket /ws/tracking mínimo :
}
}

```

Response 404 Not Found

HTTP/1.1 404 Not Found

Content-Type: application/json

```

{
  "error": "no_assigned_trip",
  "error_description": "No trip assigned for today. Contact operations.",
  "timestamp": "2026-05-20T10:30:00Z"
}

```

6.8.2 PATCH /api/v1/trips/{tripId}/boarding

Purpose: Driver marks students as boarded or no-show.

Authentication: Bearer token required (role: DRIVER).

Rate Limit: 50 requests per minute per driver.

Request

PATCH /api/v1/trips/550e8400-e29b-41d4-a716-446655440000/boarding HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Content-Type: application/json

```
{
  "accion": "MARCAR_ABORDADO",
  "reservacionId": "760e8400-e29b-41d4-a716-446655442222",
  "timestamp": 1234567890000
}
```

VALIDATIONS:

- tripId: must be valid UUID
- accion: enum [MARCAR_ABORDADO, MARCAR_NO_SHOW, DESMARCAR]
- reservacionId: must be valid UUID
- timestamp: if provided, cannot be > now()
- User.role must be DRIVER
- User must be assigned driver of tripId
- Reservation must belong to tripId

PROCESSING BY ACTION:

MARCAR_ABORDADO:

1. Change Reservation.estado to BOARDED
2. Register horaAbordaje = timestamp
3. Increment TrustScore user +2 (trip completed when disembarking)
4. Update occupancy statistics

MARCAR_NO_SHOW:

1. Change Reservation.estado to NO_SHOW
2. Register detection timestamp
3. Decrement TrustScore user -10 (not showing up)
4. Release Seat (change to AVAILABLE)

DESMARCAR:

1. If Reservation.estado == BOARDED: change to RESERVED (reverse)
2. Clear horaAbordaje

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "reservacion": {
    "id": "760e8400-e29b-41d4-a716-446655442222",
    "estudianteId": "950e8400-e29b-41d4-a716-446655444444",
    "asientoNumero": 5,
    "estado": "BOARDED",
    "horaAbordaje": "2026-05-20T15:58:00Z",
    "observaciones": null
  },
  "estadoViaje": {
    "totalAbordados": 9,
    "totalNoShows": 1,
    "totalReservaciones": 10,
    "porcentajeOcupacion": 0.60
  }
}
```

Response 404 Not Found

HTTP/1.1 404 Not Found

Content-Type: application/json

```
{
  "error": "reservacion_not_found",
  "error_description": "Reservation with ID {reservacionId} not found for this trip"
}
```

Response 409 Conflict

HTTP/1.1 409 Conflict

Content-Type: application/json

```
{
  "error": "invalid_state_transition",
  "error_description": "Cannot transition reservation from RESERVED to NO_SHOW after 3",
  "timestamp": "2026-05-20T10:30:00Z"
}
```

CASES:

- Already marked as boarded
- Excessive time to mark as no-show (> 30 min from departure)
- Reservation already cancelled

6.9 Notifications Module: /api/v1/notifications

6.9.1 POST /api/v1/notifications/register-device

Purpose: Register FCM token of mobile device.

Authentication: Bearer token required.

Rate Limit: 10 requests per minute per user.

Request

POST /api/v1/notifications/register-device HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Content-Type: application/json

```
{
  "fcmToken": "eAz...xV5Q:APA91bEd...abc123def456",
  "platform": "ANDROID"
}
```

VALIDATIONS:

- fcmToken: cannot be empty; must have valid FCM token format (150-200 chars)
- platform: enum [IOS, ANDROID]

PROCESSING:

1. Validate JWT
2. Check if token already registered for user
3. If exists: update last activity timestamp
4. If not exists: create new DeviceToken record
5. Save to DB with fecha_registro = now()

Response 201 Created

HTTP/1.1 201 Created

Content-Type: application/json

```
{
  "id": "c50e8400-e29b-41d4-a716-446655777777",
  "usuarioId": "950e8400-e29b-41d4-a716-446655444444",
}
```

```
"platform": "ANDROID",
"estado": "active",
"fechaRegistro": "2026-05-20T10:30:00Z"
}
```

Response 400 Bad Request

HTTP/1.1 400 Bad Request
Content-Type: application/json

```
{
  "error": "invalid_fcm_token",
  "error_description": "FCM token format is invalid or too short"
}
```

6.9.2 PUT /api/v1/notifications/preferences

Purpose: Update user notification preferences.

Authentication: Bearer token required.

Request

PUT /api/v1/notifications/preferences HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...
Content-Type: application/json

```
{
  "notificacionesBusSaliendo": true,
  "notificacionesBusProximo": true,
  "notificacionesReservaConfirmada": true,
  "notificacionesCancelacion": false,
  "notificacionesPuntosConfianza": true
}
```

VALIDATIONS:

- All fields must be boolean

Response 200 OK

HTTP/1.1 200 OK
Content-Type: application/json

```
{
```

```
"id": "d50e8400-e29b-41d4-a716-446655888888",
"usuarioId": "950e8400-e29b-41d4-a716-446655444444",
"notificacionesBusSaliendo": true,
"notificacionesBusProximo": true,
"notificacionesReservaConfirmada": true,
"notificacionesCancelacion": false,
"notificacionesPuntosConfianza": true,
"fechaActualizacion": "2026-05-20T10:30:00Z"
}
```

6.10 Administration Module: /api/v1/admin

6.10.1 POST /api/v1/admin/routes

Purpose: Create new transportation route.

Authentication: Bearer token required (role: ADMIN).

Rate Limit: 10 requests per minute per admin.

Request

POST /api/v1/admin/routes HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Content-Type: application/json

```
{
  "nombre": "Circunvalación Campus Principal",
  "descripcion": "Ruta nocturna que recorre el perímetro del campus",
  "paradas": [
    {
      "nombre": "Entrada Principal",
      "orden": 1,
      "latitud": -0.2156,
      "longitud": -78.4678
    },
    {
      "nombre": "Biblioteca Central",
      "orden": 2,
      "latitud": -0.2170,
      "longitud": -78.4690
    }
  ]
}
```

VALIDATIONS:

- nombre: cannot be empty; maximum 255 characters; must be unique
- descripcion: maximum 1000 characters
- paradas: minimum 2 stops; maximum 10 stops
- Each stop:
 - * nombre: non-empty; maximum 100 characters
 - * orden: integer, no duplicates
 - * latitud: valid range (-90 to 90)
 - * longitud: valid range (-180 to 180)

Response 201 Created

HTTP/1.1 201 Created

Location: /api/v1/admin/routes/150e8400-e29b-41d4-a716-446655446666

Content-Type: application/json

```
{
  "id": "150e8400-e29b-41d4-a716-446655446666",
  "nombre": "Circunvalación Campus Principal",
  "descripcion": "Ruta nocturna que recorre el perímetro del campus",
  "activa": true,
  "paradas": [
    {
      "id": "250e8400-e29b-41d4-a716-446655447777",
      "nombre": "Entrada Principal",
      "orden": 1,
      "latitud": -0.2156,
      "longitud": -78.4678
    }
  ],
  "fechaCreacion": "2026-05-20T10:30:00Z"
}
```

Response 400 Bad Request

HTTP/1.1 400 Bad Request

Content-Type: application/json

```
{
  "error": "validation_error",
  "error_description": "Route name must be unique",
  "details": ["nombre"]
}
```

```
}
```

6.10.2 POST /api/v1/admin/trips

Purpose: Create new scheduled trip.

Authentication: Bearer token required (role: ADMIN).

Request

POST /api/v1/admin/trips HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Content-Type: application/json

```
{
  "busId": "050e8400-e29b-41d4-a716-446655445555",
  "routeId": "150e8400-e29b-41d4-a716-446655446666",
  "conductorId": "450e8400-e29b-41d4-a716-446655449999",
  "fechaHoraSalida": "2026-05-20T16:00:00Z"
}
```

VALIDATIONS:

- busId: must exist in DB
- routeId: must exist in DB
- conductorId: must exist and be DRIVER
- fechaHoraSalida: cannot be in the past; maximum 30 days in future

PROCESSING:

1. Create Trip with status = SCHEDULED
2. Create Seat records (one per bus seat)
3. All Seats with status = AVAILABLE initially

Response 201 Created

HTTP/1.1 201 Created

Location: /api/v1/admin/trips/550e8400-e29b-41d4-a716-446655440000

Content-Type: application/json

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "busId": "050e8400-e29b-41d4-a716-446655445555",
  "routeId": "150e8400-e29b-41d4-a716-446655446666",
  "conductorId": "450e8400-e29b-41d4-a716-446655449999",
}
```

```
"fechaHoraSalida": "2026-05-20T16:00:00Z",
"estado": "SCHEDULED",
"asientosDisponibles": 15,
"asientosTotales": 15,
"fechaCreacion": "2026-05-20T10:30:00Z"
}
```

6.10.3 PATCH /api/v1/admin/users/{id}/trust-score

Purpose: Manually adjust user trust score.

Authentication: Bearer token required (role: ADMIN).

Request

```
PATCH /api/v1/admin/users/950e8400-e29b-41d4-a716-446655444444/trust-score HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...
Content-Type: application/json
```

```
{
  "cambio": -10,
  "razon": "Comportamiento inapropiado en viaje",
  "temporal": false,
  "diasExpiracion": null
}
```

VALIDATIONS:

- cambio: integer, range -100 to +100
- razon: string, maximum 500 characters
- temporal: boolean (if true, specify diasExpiracion)
- diasExpiracion: integer, if temporal == true

PROCESSING:

1. Get current TrustScore
2. Calculate newScore = current + cambio
3. Validate not out of range [0, 100]
4. If temporal: create penalty with expiration date
5. Record change in history
6. Emit TrustScoreAdjustedEvent

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "usuarioId": "950e8400-e29b-41d4-a716-446655444444",
  "puntuacionAnterior": 85,
  "puntuacionNueva": 75,
  "cambio": -10,
  "razon": "Comportamiento inapropiado en viaje",
  "penalizacion": {
    "id": "e50e8400-e29b-41d4-a716-446655999999",
    "tipo": "MISCONDUCT",
    "fechaInicio": "2026-05-20T10:30:00Z",
    "fechaExpiracion": null,
    "estado": "ACTIVE"
  },
  "fechaActualizacion": "2026-05-20T10:30:00Z"
}
```

6.11 Supervision Module: /api/v1/supervisor

6.11.1 GET /api/v1/supervisor/all-locations

Purpose: Snapshot of current location of all active buses.

Authentication: Bearer token required (role: SUPERVISOR).

Rate Limit: 60 requests per minute per supervisor.

Request

```
GET /api/v1/supervisor/all-locations HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...
```

Response 200 OK

```
HTTP/1.1 200 OK
Content-Type: application/json
```

```
{
  "timestamp": 1234567890000,
  "totalBusesActivos": 8,
  "buses": [
    {
      "busId": "050e8400-e29b-41d4-a716-446655445555",
```



```

    "numeroPlaca": "UIO-1234",
    "latitude": -0.2156,
    "longitude": -78.4678,
    "accuracy": 8.5,
    "velocity": 45.2,
    "tripId": "550e8400-e29b-41d4-a716-446655440000",
    "routeName": "Circunvalación Campus",
    "conductorId": "450e8400-e29b-41d4-a716-446655449999",
    "conductorNombre": "Juan Pérez García",
    "asientosOcupados": 8,
    "asientosTotales": 15,
    "proximaParada": {
      "nombre": "Biblioteca Central",
      "orden": 2,
      "distanciaMetros": 450
    },
    "incidentes": [
      {
        "id": "f50e8400-e29b-41d4-a716-446655aaaaaa",
        "tipo": "delay",
        "descripcion": "Tráfico en calle 10 de Agosto",
        "severidad": "MEDIA",
        "timestamp": 1234567880000
      }
    ]
  },
  {
    "busId": "150e8400-e29b-41d4-a716-446655bbbbbbb",
    "numeroPlaca": "UIO-5678",
    // ... next bus
  }
]
}

```

FIELDS:

- busId: unique bus identifier
- numeroPlaca: visible license plate
- latitude/longitude: current coordinates (from Redis)
- accuracy: GPS precision in meters
- velocity: current speed in km/h
- tripId: trip currently operating
- routeName: route name

- conductorId/conductorNombre: driver identification
- asientosOcupados: number of students boarded
- asientosTotales: bus capacity
- proximaParada: next stop on trip
- incidentes: list of open incidents for this trip

6.11.2 GET /api/v1/supervisor/statistics

Purpose: Operating system statistics for current day.

Authentication: Bearer token required (role: SUPERVISOR).

Request

GET /api/v1/supervisor/statistics HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "timestamp": 1234567890000,
  "fecha": "2026-05-20",
  "operaciones": {
    "totalBusesScheduled": 10,
    "totalBusesActivos": 8,
    "totalBusesRetrasados": 2,
    "totalBusesEnMantenimiento": 0
  },
  "estudiantes": {
    "totalReservas": 95,
    "totalAbordo": 87,
    "totalNoShows": 3,
    "totalCancelaciones": 5
  },
  "rendimiento": {
    "tasaOcupacionPromedio": 0.72,
    "delayPromedio": "4 minutos",
    "delayMaximo": "12 minutos",
    "puntualidadPorcentaje": 0.80
  },
}
```

```
"incidentes": {
  "incidentosAbiertos": 3,
  "incidentosCerrados": 1,
  "tiposIncidentes": [
    {
      "tipo": "delay",
      "cantidad": 2
    },
    {
      "tipo": "mechanical_issue",
      "cantidad": 1
    }
  ]
},
"disponibilidad": {
  "asientosDisponiblesTotal": 45,
  "asientosReservadosTotal": 132,
  "asientosOcupadosTotal": 87
},
"conductores": {
  "conductoresActivos": 8,
  "conductoresSuspendidos": 0,
  "promedioCalificacion": 4.7
}
}
```

NOTES:

- Statistics are recalculated every minute
- Delay calculated by comparing scheduled time vs current location
- Average occupancy = total students boarded / total capacity

6.11.3 GET /api/v1/supervisor/incidents

Purpose: Get report of today's open incidents.

Authentication: Bearer token required (role: SUPERVISOR).

Request

GET /api/v1/supervisor/incidents?estado=ABIERTO&tipo=all HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

QUERY PARAMETERS:

- estado: enum [ABIERTO, CERRADO, ALL] (default: ABIERTO)
- tipo: enum [delay, mechanical_issue, misconduct, safety, all] (default: all)

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "total": 3,
  "incidentes": [
    {
      "id": "f50e8400-e29b-41d4-a716-446655aaaaaa",
      "tripId": "550e8400-e29b-41d4-a716-446655440000",
      "busId": "050e8400-e29b-41d4-a716-446655445555",
      "tipo": "delay",
      "severidad": "MEDIA",
      "descripcion": "Tráfico en calle 10 de Agosto",
      "estado": "ABIERTO",
      "delayMinutos": 12,
      "fechaCreacion": "2026-05-20T16:05:00Z",
      "reportadoPor": "Sistema automático",
      "notas": [
        {
          "timestamp": "2026-05-20T16:10:00Z",
          "autor": "Carlos Supervisor",
          "contenido": "Monitorear situación de tráfico"
        }
      ]
    }
  ]
}
```

6.11.4 POST /api/v1/supervisor/incidents/{id}/note

Purpose: Add note to incident (audit).

Authentication: Bearer token required (role: SUPERVISOR).

Request

POST /api/v1/supervisor/incidents/f50e8400-e29b-41d4-a716-446655aaaaaa/note HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Content-Type: application/json

```
{
  "contenido": "Contactado al conductor. Tráfico despejándose.",
  "timestamp": 1234567890000
}
```

VALIDATIONS:

- contenido: cannot be empty; maximum 1000 characters
- timestamp: optional; if not provided, use now()

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "id": "f50e8400-e29b-41d4-a716-446655aaaaaa",
  "notasTotal": 2,
  "ultimaNota": {
    "timestamp": "2026-05-20T16:15:00Z",
    "autor": "Carlos Supervisor",
    "contenido": "Contactado al conductor. Tráfico despejándose."
  }
}
```

6.11.5 POST /api/v1/supervisor/trips/{id}/cancel

Purpose: Cancel trip due to emergency or force majeure.

Authentication: Bearer token required (role: SUPERVISOR or ADMIN).

Request

POST /api/v1/supervisor/trips/550e8400-e29b-41d4-a716-446655440000/cancel HTTP/1.1

Host: api.buslink.uce.edu.ec

Authorization: Bearer eyJhbGc...

Content-Type: application/json

```
{
  "razon": "Falla mecánica en bus",
  "notificar": true
}
```

VALIDATIONS:

- razon: string, non-empty, maximum 500 characters
- notificar: boolean (if true, send FCM to students)

PROCESSING:

1. Change Trip.estado to CANCELLED
2. For each Reservation on trip:
 - Change to CANCELLED_BY_ADMIN
 - Release Seat (change to AVAILABLE)
3. If notificar == true:
 - Send FCM to all students with reservation
 - Message: "Your trip has been cancelled. Reason: {razon}"
4. Emit TripCancelledEvent
5. Register in audit log

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "estado": "CANCELLED",
  "razon": "Falla mecánica en bus",
  "fechaCancelacion": "2026-05-20T16:20:00Z",
  "estudiantesNotificados": 8,
  "reservacionesCanceladas": 8
}
```

6.11.6 GET /api/v1/supervisor/audit-logs

Purpose: Get audit logs (administrative changes).

Authentication: Bearer token required (role: SUPERVISOR or ADMIN).

Request

```
GET /api/v1/supervisor/audit-logs?desde=2026-05-20&hasta=2026-05-20&\
recurso=TRIP&pagina=1&tamano=50 HTTP/1.1
Host: api.buslink.uce.edu.ec
Authorization: Bearer eyJhbGc...
```

QUERY PARAMETERS:

- desde: ISO 8601 date (default: today)

- hasta: ISO 8601 date (default: today)
- recurso: enum [USER, TRIP, ROUTE, BUS, RESERVATION, ALL] (default: ALL)
- accion: enum [CREATE, UPDATE, DELETE, SUSPEND, CANCEL, ALL] (default: ALL)
- usuario: UUID of user who performed action (optional)
- pagina: integer (default: 1)
- tamaño: integer (default: 50, maximum: 500)

Response 200 OK

HTTP/1.1 200 OK

Content-Type: application/json

```
{
  "total": 24,
  "pagina": 1,
  "tamaño": 50,
  "logs": [
    {
      "id": "g50e8400-e29b-41d4-a716-446655bbbbbb",
      "timestamp": "2026-05-20T16:20:00Z",
      "usuario": {
        "id": "050e8400-e29b-41d4-a716-446655aaaaaa",
        "email": "admin@uce.edu.ec",
        "nombre": "Carlos Administrador"
      },
      "accion": "CANCEL",
      "recurso": "TRIP",
      "recursoId": "550e8400-e29b-41d4-a716-446655440000",
      "detalles": {
        "razon": "Falla mecánica en bus",
        "estudiantesAfectados": 8
      },
      "ipOrigen": "192.168.1.100",
      "userAgent": "Mozilla/5.0..."
    }
  ]
}
```

AUDIT FIELDS:

- All administrative changes are logged
- Includes origin IP and User-Agent
- Immutable (cannot be edited or deleted)
- Required for compliance and debugging

6.12 Global Error Handling and HTTP Codes

6.12.1 Standardized HTTP Codes

2xx Success:

- 200 OK: Successful operation (response with body)
- 201 Created: Resource created successfully
 - Location Header: URL of new resource
- 204 No Content: Successful operation (no response body)

4xx Client Error:

- 400 Bad Request: Request validation failed
- 401 Unauthorized: Token absent or invalid
- 403 Forbidden: Insufficient role or permission denied
- 404 Not Found: Resource does not exist
- 409 Conflict: Inconsistent state (race condition)
- 422 Unprocessable Entity: Business validation failed
- 429 Too Many Requests: Rate limiting limit exceeded
 - Headers: Retry-After, X-RateLimit-Reset

5xx Server Error:

- 500 Internal Server Error: Unexpected error
 - Include traceId for debugging
- 502 Bad Gateway: Proxy/load balancer error
- 503 Service Unavailable: Maintenance or overload
- 504 Gateway Timeout: Backend service timeout

6.12.2 Standard Error Structure

HTTP/1.1 {statusCode} {reasonPhrase}

Content-Type: application/json

```
{
  "error": "error_code_snake_case",
  "error_description": "Human-readable error description",
  "timestamp": "2026-05-20T10:30:00Z",
  "traceId": "a1b2c3d4e5f6 (optional, only in 5xx)",
  "data": {
    // Additional contextual information
  }
}
```

FIELDS:

- error: programmatic code (enum, does not vary)
- error_description: human-readable message (may vary by context)
- timestamp: when error occurred
- traceId: unique identifier for correlating with logs
- data: additional context (e.g. current rate limit)

6.12.3 Rate Limit Headers

X-RateLimit-Limit: 10 (maximum requests allowed)

X-RateLimit-Remaining: 3 (remaining requests in window)

X-RateLimit-Reset: 1234567890 (timestamp when counter resets)

Retry-After: 60 (seconds to wait before retrying, on 429)

6.13 WebSocket: Supervisor Channel (All-Buses Broadcast)

6.13.1 Subscription to Supervisor Channel

Fundamental difference: The supervisor channel receives ALL buses simultaneously, unlike the student who only receives their reserved trip.

Handshake

GET /ws/tracking HTTP/1.1

Host: api.buslink.uce.edu.ec

Upgrade: websocket

Connection: Upgrade

Sec-WebSocket-Key: dGhlIHhnbXBsZSBub25jZQ==

Sec-WebSocket-Version: 13

Authorization: Bearer {supervisorJWT}

X-Client-Type: electron-supervisor

VALIDATIONS:

1. Valid JWT and role == SUPERVISOR
2. X-Client-Type == electron-supervisor (informative for logging)
3. If both OK: connects to /topic/supervisor/all-buses

RESPONSE:

HTTP/1.1 101 Switching Protocols

Upgrade: websocket

Connection: Upgrade

Sec-WebSocket-Accept: s3pPLMBiTxaQ9kYGzzhZRbK+x0o=

All-Buses Update Message (Backend → Supervisor)**Frequency:** Every 2-5 seconds (configurable)**Structure:**

```
{
  "type": "all_buses_update",
  "timestamp": 1234567890000,
  "buses": [
    {
      "busId": "050e8400-e29b-41d4-a716-446655445555",
      "numeroPlaca": "UI0-1234",
      "latitude": -0.2156,
      "longitude": -78.4678,
      "accuracy": 8.5,
      "velocity": 45.2,
      "tripId": "550e8400-e29b-41d4-a716-446655440000",
      "routeName": "Circunvalación Campus",
      "conductorId": "450e8400-e29b-41d4-a716-446655449999",
      "conductorNombre": "Juan Pérez García",
      "asientosOcupados": 8,
      "asientosTotales": 15,
      "proximaParada": {
        "nombre": "Biblioteca Central",
        "orden": 2,
        "distanciaMetros": 450,
        "etaSegundos": 54
      },
    },
    "incidentes": [
      {
        "id": "f50e8400-e29b-41d4-a716-446655aaaaaa",
        "tipo": "delay",
        "descripcion": "Tráfico en calle 10 de Agosto",
        "severidad": "MEDIA",
        "timestamp": 1234567880000
      }
    ],
    "estado": "ACTIVE",
    "ultimaActualizacion": 1234567890000
  },
  {
    "busId": "150e8400-e29b-41d4-a716-446655bbbbbbb",
    // ... next bus
  }
}
```

```
    }
  ],
  "statisticsSnapshot": {
    "totalBusesActivos": 8,
    "totalEstudiantesAbordo": 95,
    "delayPromedioMinutos": 4,
    "incidentesTotales": 3,
    "asientosDisponiblesRed": 45,
    "tasaOcupacionPromedio": 0.72,
    "busesRetrasados": 2,
    "busesEnMantenimiento": 0
  },
  "metadata": {
    "serverTime": 1234567890000,
    "secuencial": 12345
  }
}
```

IMPORTANT NOTES:

- Each message contains COMPLETE state of all buses
- Not delta/diff; it is a complete snapshot
- Supervisor can use secuencial to detect lost messages
- etaSegundos is dynamically calculated based on current velocity
- Incidents are only the open ones for that trip

6.14 Pagination Conventions

STANDARD PARAMETERS:

- pagina: integer, minimum 1 (default: 1)
- tamaño: integer, minimum 1, maximum specified by endpoint (default: 20)

PAGINATED RESPONSE:

```
{
  "total": 150 (total records without pagination),
  "pagina": 1,
  "tamaño": 20,
  "totalPaginas": 8,
  "items": [...]
}
```

NAVIGATION:

- If pagina == 1: no previous page

- If `pagina * tamano >= total`: no next page
- Client can calculate: `proximaPagina = pagina + 1` (if `< totalPaginas`)

URL EXAMPLES:

GET `/api/v1/reservations?pagina=1&tamano=20`

GET `/api/v1/admin/audit-logs?pagina=3&tamano=100`

6.15 Recommended Security Headers

All endpoints must return these headers:

`Content-Security-Policy: default-src 'self'; script-src 'self'; style-src 'self' 'unsafe-inline'`

`X-Content-Type-Options: nosniff`

`X-Frame-Options: DENY`

`X-XSS-Protection: 1; mode=block`

`Strict-Transport-Security: max-age=31536000; includeSubDomains; preload`

`Access-Control-Allow-Origin: {valid origin}`

`Access-Control-Allow-Credentials: true`

Chapter 7

Data Model (Entity-Relationship)

7.1 Primary Entities

7.1.1 User

- `id` (UUID, PK)
- `email` (unique, @uce.edu.ec)
- `nombres`, `apellidos`
- `rol` (STUDENT, DRIVER, ADMIN, SUPERVISOR)
- `estado` (ACTIVE, INACTIVE, SUSPENDED)
- `google_id`
- `fecha_primer_login`, `fecha_ultimo_login`

7.1.2 Route

- `id` (UUID, PK)
- `nombre`, `descripcion`
- `activa` (boolean)

7.1.3 Stop

- `id` (UUID, PK)
- `ruta_id` (FK → Route)
- `orden_secuencial`, `nombre`
- `latitud`, `longitud`, `horario_llegada_estimado`

7.1.4 Bus

- id (UUID, PK)
- numero_placa (unique)
- capacidad_asientos
- estado (OPERATIONAL, MAINTENANCE, DECOMMISSIONED)

7.1.5 Trip

- id (UUID, PK)
- bus_id (FK → Bus)
- ruta_id (FK → Route)
- conductor_id (FK → User)
- fecha_hora_salida
- estado (SCHEDULED, ONGOING, COMPLETED, CANCELLED)
- version (for optimistic locking)

7.1.6 Seat

- id (UUID, PK)
- viaje_id (FK → Trip)
- numero_asiento
- estado (AVAILABLE, RESERVED, BOARDED, CANCELLED)
- version
- Constraint: UNIQUE(viaje_id, numero_asiento)

7.1.7 Reservation

- id (UUID, PK)
- usuario_id (FK → User)
- viaje_id (FK → Trip)
- asiento_id (FK → Seat)
- fecha_creacion, fecha_cancelacion (nullable)

- `estado` (ACTIVE, CANCELLED_BY_STUDENT, CANCELLED_BY_ADMIN, NO_SHOW)
- `version`
- Constraint: UNIQUE(`viaje_id`, `usuario_id`)

7.2 Critical Indexes for Performance

- `idx_user_email` on User(email)
- `idx_trip_route_fecha` on Trip(`ruta_id`, `fecha_hora_salida`)
- `idx_seat_trip_estado` on Seat(`viaje_id`, `estado`)
- `idx_reservation_usuario` on Reservation(`usuario_id`)
- `idx_location_bus_timestamp` on TripLocationHistory(`bus_id`, `timestamp` DESC)

Chapter 8

Persistence Strategy and Migrations

8.1 Schema Versioning with Flyway

8.1.1 Problem with Hibernate DDL-Auto in Production

The `spring.jpa.hibernate.ddl-auto=update` configuration automatically generates `ALTER TABLE` statements when the application starts. This behavior is unacceptable in production due to the following risks:

- Schema alterations without version control or review
- Inability to rollback if a change breaks the database
- Possible data loss if Hibernate decides to drop columns
- Locks on tables during `ALTER TABLE` that block normal operation
- Absence of audited history of changes

8.1.2 Flyway as Project Standard

Flyway manages database changes as versioned, immutable SQL scripts. When the application starts, Flyway:

1. Queries the internal `FLYWAY_SCHEMA_HISTORY` table
2. Identifies which scripts have already been applied
3. Executes in sequential order any new scripts
4. Records each successful execution
5. If a script fails, stops the application startup

Naming Convention

V{number}__{Description}.sql, where the number is sequential and the description describes the change in snake_case.

Immutability Principle

Once a script is executed in any environment (development, staging, production), it cannot be modified. Any correction or additional change must be a new script with a higher number.

8.1.3 Defined Migration Scripts

- **V1__Initial_schema.sql:** Creates the base system tables with all foreign keys, unique constraints, and version field for optimistic locking. Also creates critical indexes for performance.
- **V2__Add_trust_score_table.sql:** Creates the trust_score table with one-to-one relationship with user and corresponding index.
- **V3__Create_location_history_table.sql:** Creates the trip_location_history table with composite index on (bus_id, timestamp DESC) for efficient historical queries.
- **V4__Add_device_tokens_and_preferences.sql:** Creates device_token and notification_preference tables.

8.2 JPA and Entity Mapping Configuration

8.2.1 N+1 Query Problem

The N+1 problem occurs when an initial query returns N records and then an additional query is executed for each record to load its relationships. The root cause is using `FetchType.EAGER` in JPA relationships.

8.2.2 Solution: Lazy Loading + Explicit Fetch

All relationships are declared with `FetchType.LAZY`. When related data is needed, the JOIN is specified in the JPQL query with the `JOIN FETCH` clause. This results in a single SQL query with the necessary JOINS, retrieving all required columns in a single network operation to the database.

Chapter 9

Resilience and Reconnection Protocols

9.1 Context: Connectivity in Quito

Quito presents particular connectivity challenges due to its mountainous topography at 2,850 meters above sea level. The identified scenarios are:

- Temporary network loss between 30 and 120 seconds (tunnels, hillside zones)
- Prolonged loss over 2 minutes (very weak coverage areas)
- Network switching (WiFi to cellular or vice versa)

9.2 Heartbeat: Zombie Connection Detection

A zombie connection is one that the client considers open but the server can no longer reach. Without heartbeat, these connections remain in the server's active sessions map consuming resources.

9.2.1 Server Mechanism (Spring WebFlux)

For each active WebSocket session, the server sends a PING frame every 30 seconds. If the send fails or the session is no longer open, the session is removed from the active sessions map.

9.2.2 Client Mechanism (React Native)

The client sends a ping-type message every 45 seconds. If the WebSocket is not in `OPEN` state when attempting to send, disconnection is detected and the reconnection process is initiated.

9.3 Reconnection with Exponential Backoff

When disconnection is detected, the client uses exponential backoff with jitter:

- Base delay of 1 second
- Each failed attempt multiplies the delay by 1.5
- Delay capped at 60 seconds maximum
- Jitter of $\pm 10\%$ to prevent thundering herd
- Maximum number of attempts: 15

9.4 Local GPS Coordinate Queue

During disconnection periods, the driver continues capturing GPS coordinates which are stored in a local in-memory queue (with AsyncStorage persistence). The queue has a maximum limit of 300 elements. Upon reconnection, the GPSSyncManager sends pending coordinates to the server including the original capture timestamp.

Chapter 10

Security Matrix

10.1 CORS Configuration

10.1.1 Allowed Origin Groups

Group 1 – React Native Mobile

- `http://localhost:19000`, `http://localhost:19001` (Expo development)
- `https://*.expo.dev` (Expo production)
- `app://buslink` (scheme-based origin)

Group 2 – Electron Desktop Development

- `http://localhost:3001`
- `http://localhost:5173` (Vite)

Group 3 – Electron Desktop Production

The Electron client uses mTLS (mutual certificates) for transport-level authentication.

10.2 SSL/TLS

In production, TLS 1.2 and 1.3 are mandatory. The SSL certificate must be issued by a trusted certificate authority (Let's Encrypt or similar). HSTS (Strict-Transport-Security) headers with `max-age` of at least one year are configured in all responses.

10.3 Rate Limiting

Implemented via a Spring MVC interceptor that applies the Token Bucket algorithm per IP for public endpoints and per authenticated user for protected endpoints.

Chapter 11

Development Schedule

This section will be completed at a later stage of the project. Sprints, phases, and delivery dates will be defined and documented here once the project team has established the detailed work plan.

Approval and Control Information

Approved by: Lead Software Architect
Effective Date: May 2026
Next Review: Upon major architectural changes or incorporation of new modules
Classification: Architecture Technical Specification (Internal)
Version: 3.0 Enterprise — Consolidation SAD v2.0 + SAD v2.5